# Yanjie Yang

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## Research Fields

Labor · Public · Experimental

Research on how public policy, household resources, and health shape household labor and financial decisions and welfare.

## Education

Ph.D. Candidate in Mathematical Economics, Texas A&M University — Expected May 2027

HEEP Fellowship Awardee

Texas A&M University — 2019–2021

M.S. in Economics

University of International Business and Economics, China — 2015–2019

B.S. in Financial Engineering

## Research Experience

Graduate Researcher, Texas A&M University — Aug 2021–Present

College Station, TX

**Selected Research Achievements**

Applied logistic regression to predict labor-market participation on 3M+ panel records; integrated web-scraped unemployment-insurance policy data into the analysis.

Estimated a monthly consumption-saving model with income and childcare-cost risk by maximum likelihood using Survey of Income and Program Participation data; evaluated payment timing and household welfare while holding annual Child Tax Credit amounts fixed.

Estimated employment and mental-health transitions in Panel Study of Income Dynamics data using maximum likelihood and Bellman-equation-based dynamic programming; presented policy simulations at AAEA 2025.

Implemented random forest and KNN for large-scale data imputation; used a Linux high-performance computing cluster to reduce computation time by 90%.

Co-designed and conducted a $125,000-funded randomized controlled trial evaluating an LLM-based nutrition recommender; collaborated on app, algorithm, and data-pipeline development.

Evaluated mental-health and vaping prediction models with random forest, gradient boosting, and cross-validation in H2O; communicated findings at APHA 2023.

## Methods and Technical Skills

Econometrics: regression analysis, difference-in-differences, instrumental variables, regression discontinuity, randomized experiments, bootstrap inference

Economic modeling: consumption-saving, labor-supply and life-cycle models; discrete choice (DCM/DDCM), simulated method of moments, maximum likelihood, dynamic programming, counterfactual analysis

Data and software: R, Python, SQL, scikit-learn, H2O, MATLAB, Excel VBA, Linux, LaTeX, EViews, Gretl; panel data, imputation, cross-validation, high-performance and parallel computing

Machine learning: logistic regression, random forest, KNN, gradient boosting, PCA, LDA, SVM

Experiments: experimental design, randomized controlled trials, A/B testing

## Job Market Paper

**When Child Benefits Arrive: Payment Timing, Liquidity, and Mothers' Welfare**

With Li Gan

October 2026 draft · Working paper updated November 5, 2026

Read the 56-page working paper (PDF)

The expanded 2021 Child Tax Credit raised benefits and paid half of them in monthly advances. Holding each family's annual credit fixed, we ask what the earlier payments were worth to married and single mothers in the Survey of Income and Program Participation. Relative to comparable women without children, mothers show no detectable change in employment and no detectable relative increase in bank balances; card debt fell, mostly among mothers who held some cash and carried debt, while low cash persisted among single mothers. A monthly consumption-saving model with income and childcare-cost risk, in which some mothers save and others spend all their cash, is estimated by maximum likelihood on each mother's records with patience set at a standard value. The model-based spender probability averages 31 percent for single mothers and 2 percent for married mothers. Relative to paying the same credit at filing, the 2021 calendar was worth $191 per married mother and $987 per single mother, and $993 and $1,820 for mothers with $1,000 or less in liquid assets. For the average married and single mother, each added credit dollar paid only at filing was worth 92 and 60 cents, against 102 and 96 cents paid as in 2021. The single-mother value is conditional on the spender description: valuing every mother as a saver gives $476, and impatient savers with conventional discount factors, valued at patient rates, give $483 to $539, about half as large.

## Working Papers

**An LLM-RAG Approach for Healthy Eating Index-Informed Personalized Food Recommendations**

With Grace Melo, Rudy Nayga, Azlan Zahid, and Yibin Wang | Under review

Develops a Healthy Eating Index-informed retrieval-augmented generation framework linking large language models to NHANES and the Food Patterns Equivalents Database. Evaluates simulated food substitutions and additions using standardized nutrition data and estimated HEI contributions.

**AI-Assisted Personalized Food Recommendations and Diet Quality: Evidence from a Randomized Controlled Trial**

With Grace Melo, Rudy Nayga, Azlan Zahid, and Yibin Wang

In a pre-registered randomized trial (AEARCTR-0015970), 839 U.S. adults received either a mobile app delivering AI-personalized meal recommendations or an otherwise identical app delivering general healthy-eating guidance. After seven weeks we find no effect on total diet quality: +0.74 HEI-2020 points (95% CI –3.49 to +4.98). Whole grains, one of thirteen components, is 1.32 points higher under treatment ($q = 0.080$). Willingness to pay rises $1.29 per standard deviation of perceived relevance ($p = 0.001$) but is flat in measured dietary change (–$0.15, $p = 0.49$). Consumers pay for how personalized an app feels, not for what it measurably does.

## Research in Progress

**Childcare Support and Maternal Employment: Income and Liquidity Channels in the UK's 2023 Reform**

With Li Gan

**Monthly or Annual? Measured Liquidity and the Welfare Economics of Child Tax Credit Disbursement Frequency**

With Li Gan

**A Life Cycle Model on Mother's Labor Participation and Mental Health**

With Di Fang and Yong Liu

Studies maternal employment and mental health using PSID panel data, a dynamic job-search model, and counterfactual policy simulations.

## Conference Presentations and Selected Papers

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| Advances with Field Experiments Conference, Chicago, IL | Sep 2026 |
| AAEA Annual Meeting, Kansas City, MO | Jul 2026 |
| AAEA Annual Meeting, Denver, CO | Jul 2025 |
| Symposium for Agricultural and Applied Economics Research, College Station, TX | May 2025 |
| American Public Health Association Annual Meeting, Atlanta, GA | Nov 2023 |
| American Academy for Health Behavior Annual Meeting, San Francisco, CA | Mar 2023 |
| American Public Health Association Annual Meeting, Boston, MA | Nov 2022 |
| Selected paper | |
| ASHEcon Annual Meeting, Minneapolis, MN | Jun 2026 |

## Teaching Experience

Texas A&M University

Co-Instructor, AGEC 317 Economic Analysis for Agribusiness Management — Spring 2023

Taught econometric methods and delivered hands-on labs in Excel; mentored student projects.

- Received positive student feedback that AGEC 317 supported summer internship work and informed career decisions.

Graduate Teaching Assistant

AGEC 447: Food and Agricultural Price Analysis — Fall 2023, 2024

Senarath Dharmasena

AGEC 453: International Agribusiness Marketing — Spring 2024

Emiliano Lopez Barrera

AGEC 448: Agricultural Commodity Futures — Spring 2022, 2024

Dimitry Vedenov

AGEC 344: Food and Agricultural Law — Spring 2022; Fall 2023

Fred Boadu

AGEC 317: Economic Analysis for Agribusiness Management — Fall 2021, 2022

Kuo Liu and Dominic Vieira

AGEC 330: Financial Management in Agriculture — Summer 2022

Chrystol Thomas

AGEC 413: Agricultural Cooperatives — Fall 2021

John Park

## Professional and Entrepreneurial Experience

Business Analyst Intern, Tesla Academy, Tesla — Dec 2018–Apr 2019

Beijing, China

Automated sales-staff assignment reporting with Excel VBA, reducing analysis time from 30 minutes to 1 minute.

Created and shared Model 3 training materials on MS Yammer to support sales-staff onboarding.

Cofounder, Chuangzhi Consulting Studio — Jan 2016–Dec 2018

University of International Business and Economics, Beijing, China

Helped a startup team secure US$1.5 million in funding from Baidu Inc. by developing its business plan.

Advised student startup teams on business planning, funding, and marketing; four teams earned first- or second-place awards in the National Innovation Competition.

Registered the studio as a company and participated in the National Innovation Training Program.

## Academic Service

Student Lead, AGEC Seminar Committee — Aug 2023–Oct 2024

Texas A&M University

Coordinated weekly academic seminars with 20+ invited speakers from across the U.S.; managed logistics, communications, speaker invitations, and a $10,000 budget for event operations and speaker accommodations.

Travel Grant Officer, AGEC Graduate Student Association — Sep 2022–Aug 2023

Texas A&M University

Shared application information and administered graduate-student travel grant budgets.

Student Volunteer, Social Media Team — Sep 2022–Dec 2023

AAEA Committee on Women in Agricultural Economics (CWAE)

Developed content highlighting women agricultural economists' achievements and conducted outreach.

## Fellowships and Travel Grants

HEEP Fellowship, Texas A&M University — 2021

Departmental AAEA Travel Grants — 2022, 2025, 2026

GSA Travel Grants — 2022, 2026

## Additional Training and Languages

Decision Making and Reinforcement Learning, Columbia University/Coursera — Aug 2025

Machine Learning, Stanford/Coursera certificate — 2022

Learning SSH; Learning Linux Command Line, LinkedIn Learning — 2022

English Language Proficiency Exam certificate, Texas A&M University — 2021

Languages: Mandarin Chinese (native), English (fluent)

## References

Li Gan
Grace Melo
Rudy Nayga